IS1203
Foundations of Information Systems

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Topic 4: Acquiring IT/IS Infrastructure Capabilities and Performance Evaluation

1. IT/IS Infrastructure
2. Business Value of IS
3. IS Infrastructure and Service Acquiring Process
4. Souring and Outsourcing IS Services and Applications
5. IT/IS Performance Evaluation and Management

Intended Learning Outcomes

- Define IT infrastructure.
- Describe the components of IS infrastructure.
- Determine the business value of IS.
- Identify the steps in IS acquisition.
- Evaluate sourcing and outsourcing options .
- Assess IT/IS infrastructure investments.
- Describe the benefits an IS can bring to a firm.

Introduction

- Growing importance of electronic infrastructure and the declining importance of physical assets and location
- More and more of a company's cash flow is online
- Longer-term decisions about IT/IS investments differentiate the company's competitive capabilities
- New IT infrastructure opportunities can support many decision-making processes

IT Infrastructure

Set of H/W and S/W required to operate entire enterprise

- Technical-IT infrastructure
  - Technology,
  - Applications, and
  - Datasets
- Human-IT infrastructure
  - Data workers and information management personnel
  - Knowledge and capabilities required to manage organizational **IT resources**

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IS infrastructure and services

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6

IS Infrastructure: Software

- Organizations have to rely on a variety of different software
- Continuously upgrading OSs and applications can be a huge cost factor (for labour and s/w)
- S/W infrastructure management using
  - open-source software,
  - software integration tools
  - application service providers

Open-Source Applications

- Source code can be modified and customized to suit users' needs.
- Typically distributed under licenses defined by the Open-Source Initiative (OSI)
- Open-source licenses: GNU General Public License (GPL), Apache, and MIT License. (Refer to https://www.gnu.org/licenses/licenses.html)
- e.g., Linux, Ubuntu, FreeBSD, Apache Web Server, Mozilla Firefox, LibreOffice, Apache OpenOffice, VLC Media Player, Audacity, Python, Ruby, WordPress, Joomla, Git, VirtualBox, OpenVPN, GIMP

Integration tools

- Integrate software products which can interoperate very well
- Aimed to improve efficiency and enhance functionality and data flow between the integrated parts
- E.g., Approaches for integration
  - API Integration e.g. Google's Custom Search JSON API to develop websites and applications to retrieve and display search results from Programmable Search Engine
  - Middleware Integration: Middleware is a product that sits between multiple primary business systems. E.g., Codeless Platforms' BPA Platform can automate manual processes such as getting data from a business application into another without the need for an expensive IT team
  - Cloud Integration: Helps manage data flow and processes across different environments. E.g. Zapier- to connect different web applications and automate workflows between them

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10

ACTIVITY

1. What is a software bug?
2. What is software asset management?

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IS Infrastructure: Hardware

- An integral part of infrastructure
  - Computers
  - Networking devices

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Hardware solutions

- Fluctuating computing needs - On-demand computing
- Large scale problems with cost or use limitations – Grid computing
- Cost issues related to bandwidth and processing speed - Edge computing
- Low reliability and high maintenance cost - Autonomic computing
- Cost, access and software selecting and updating issues - Cloud Computing

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On-demand computing

- Allocate available resources based on users' needs (on a pay-per-use basis)
- Rent resources from an external provider- Utility computing
  - Rent resources for processing, data storage, or networking, on an as-needed basis
  - Tenant receives a monthly bill for the services used
  - Service provider should do the tasks such as managing, maintaining, and upgrading the infrastructure
  - All charges are added to the utility bill

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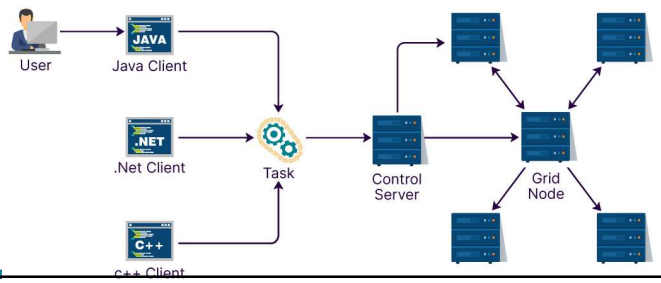
Grid computing

- To overcome cost or use limitations
  - Supercomputers are very expensive, cannot afford or justify
  - Some tasks are beyond the capacity of available supercomputers
  - Can solve very large-scale problems as well as multiple smaller problems
  - Dedicated grids
    - To perform the grid's computing tasks and
    - overcome speed issues



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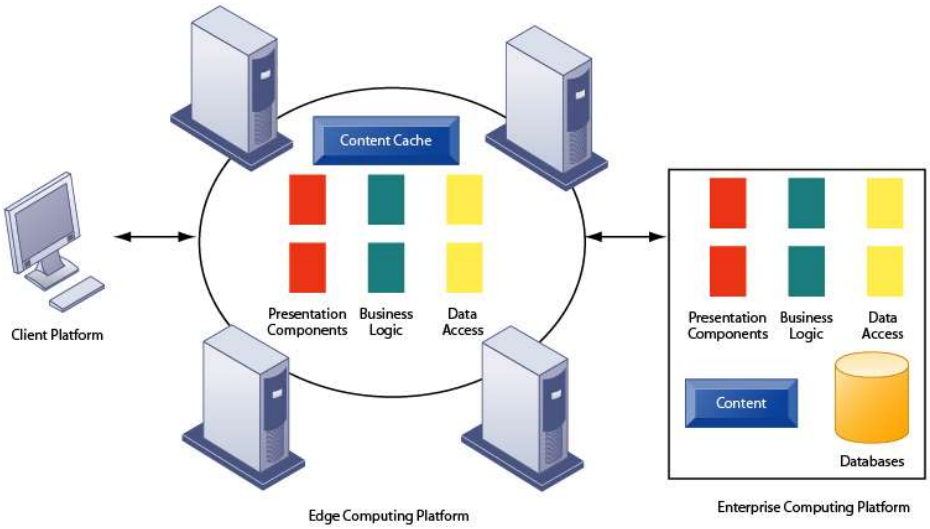
Edge computing

For bandwidth and processing speed issues



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Autonomic computing

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Green Computing

- Minimize the negative impact on the environment
- Reduce power consumption
- Adopt better practices and technologies for designing, developing, using and disposing of hardware
- Green computing approaches:
  - Green Use: e.g., virtualization
  - Green Disposal: Reuse, recycle or properly dispose
  - Green Design: Designing energy-efficient and effective systems
  - Green Manufacturing: Use Biodegradable materials, manufacture long usable, recyclable products and reduce wastage

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Virtual Computing

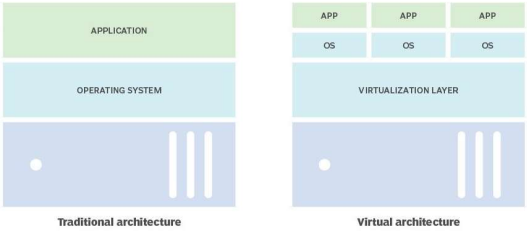
Enables running multiple operating systems on a single computer by virtualizing CPU, storage and RAM.

Key technology involved in Virtualization

1. CPU virtualization technology
2. Memory virtualization technology
3. Virtualization technology of equipment, I/O, and network ports
4. Real-time virtual migration technology

Advantages

1. Integrate resources
2. Low energy consumption
3. Reduce operating costs
4. More control on Applications



https://linux.how2shout.com/what-is-virtualization-technology-and-its-advantages/

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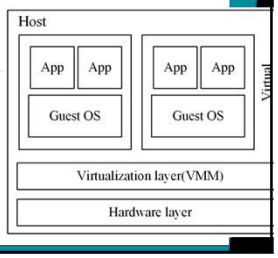
Top VM Software Download for Windows, Linux & Mac

Name	Platform	Free Trial
Virtualization Manager	Windows, Mac, Linux	30-Days Free Trial
VM Monitor	Windows, Mac, Linux	30-Days Free Trial
Site24x7's VM Monitoring	iOS, Android, Windows, Mac, and Linux	30-day free trial
VMware Fusion (Mac)	Mac	30-Days Free Trial
Parallels	Mac	14-Days Free Trial



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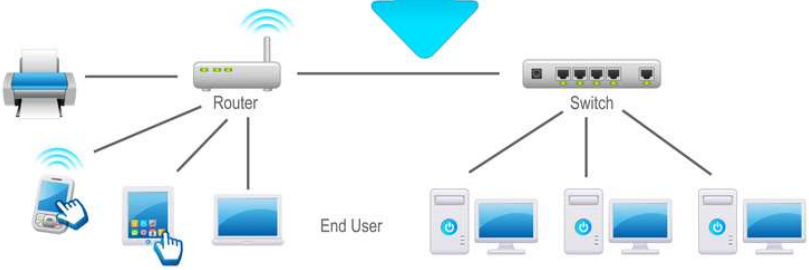
On-demand access, via the internet, to computing resources



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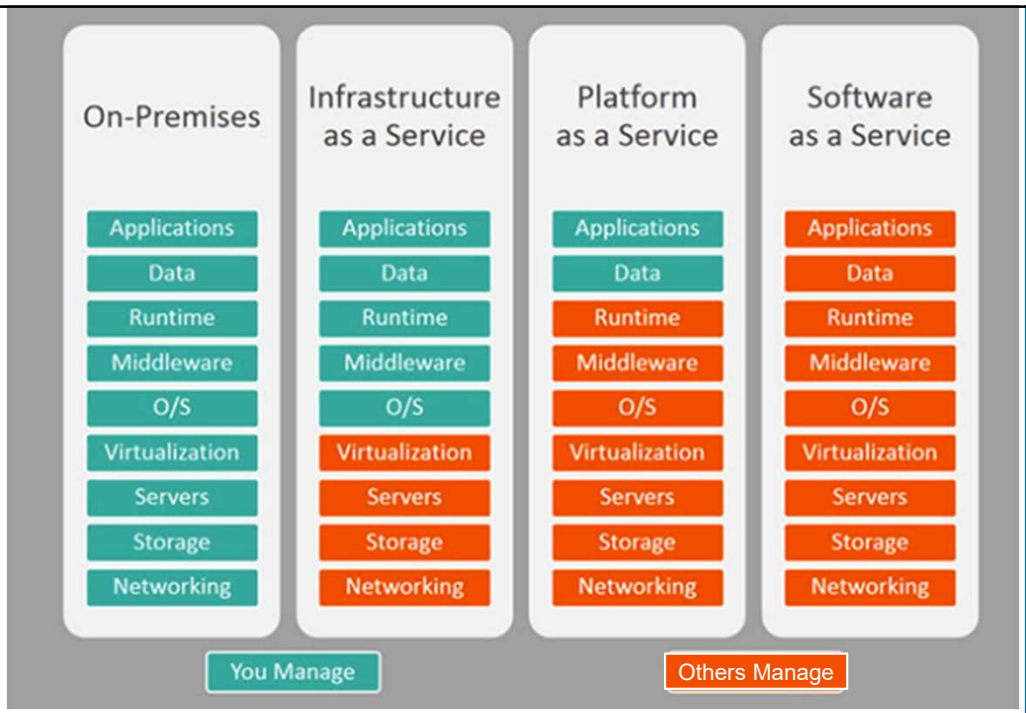

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Cloud Computing Requirements

PaaS: Develop, run, and manage app interfaces and microservices, Set up a data warehouse and use advanced analytical techniques

SaaS: Live events such as eSports, activities which need quick, easy, and affordable solutions.

IaaS: Meet requirements for supercomputers and clusters, For start-ups when unsure about new applications' demands

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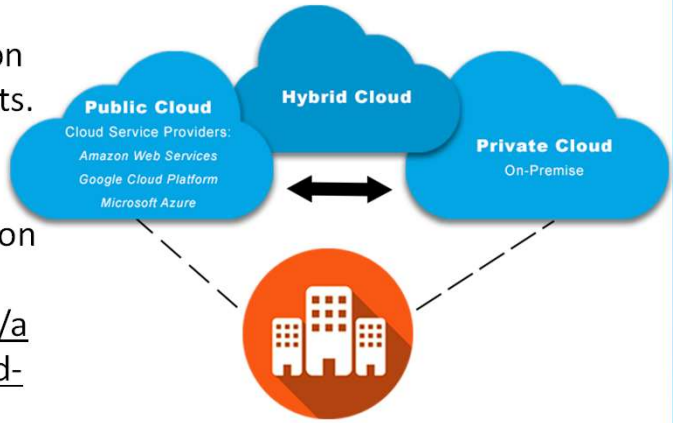
Hybrid Cloud Computing

- What is "Hybrid Cloud"?
- **How Hybrid Cloud Computing is Beneficial for Entrepreneurs?**

"Can optimize the systems based on the company's unique requirements. Helps to accelerate the delivery of services and applications"

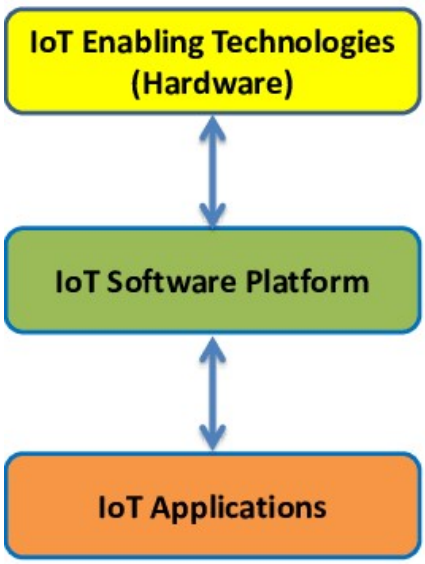
Madelyn Wilson

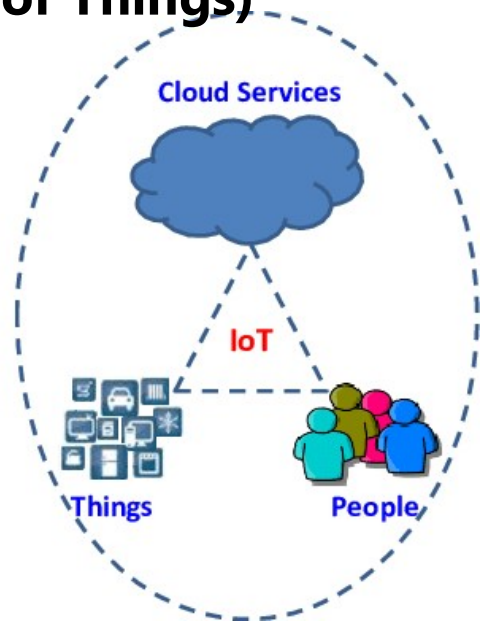
Refer to https://www.digitaldoughnut.com/articles/2018/july/how-hybrid-cloud-computing-is-beneficial



42

IoT (Internet of Things)





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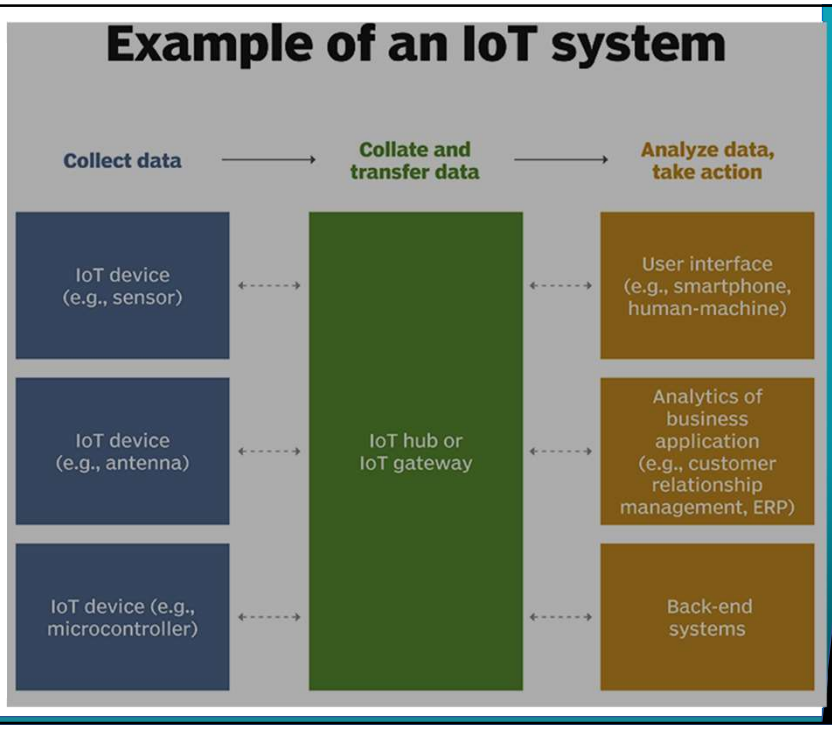
IoT to;

- monitor overall business processes,
- improve customer experience,
- save time and money,
- enhance employee productivity,
- integrate and adapt business models,
- make better business decisions, and
- generate more revenue.



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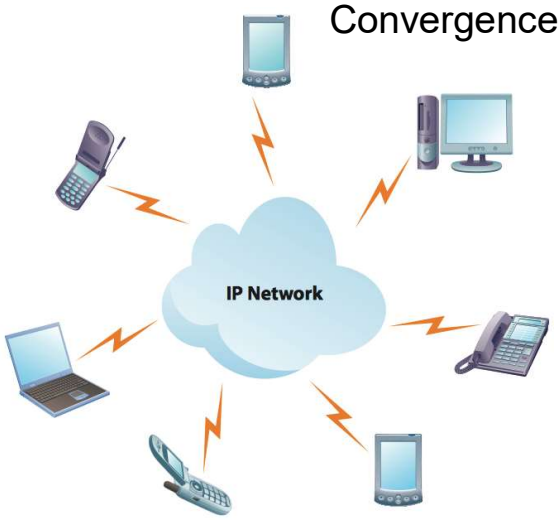
IS Infrastructure: Communication and Collaboration

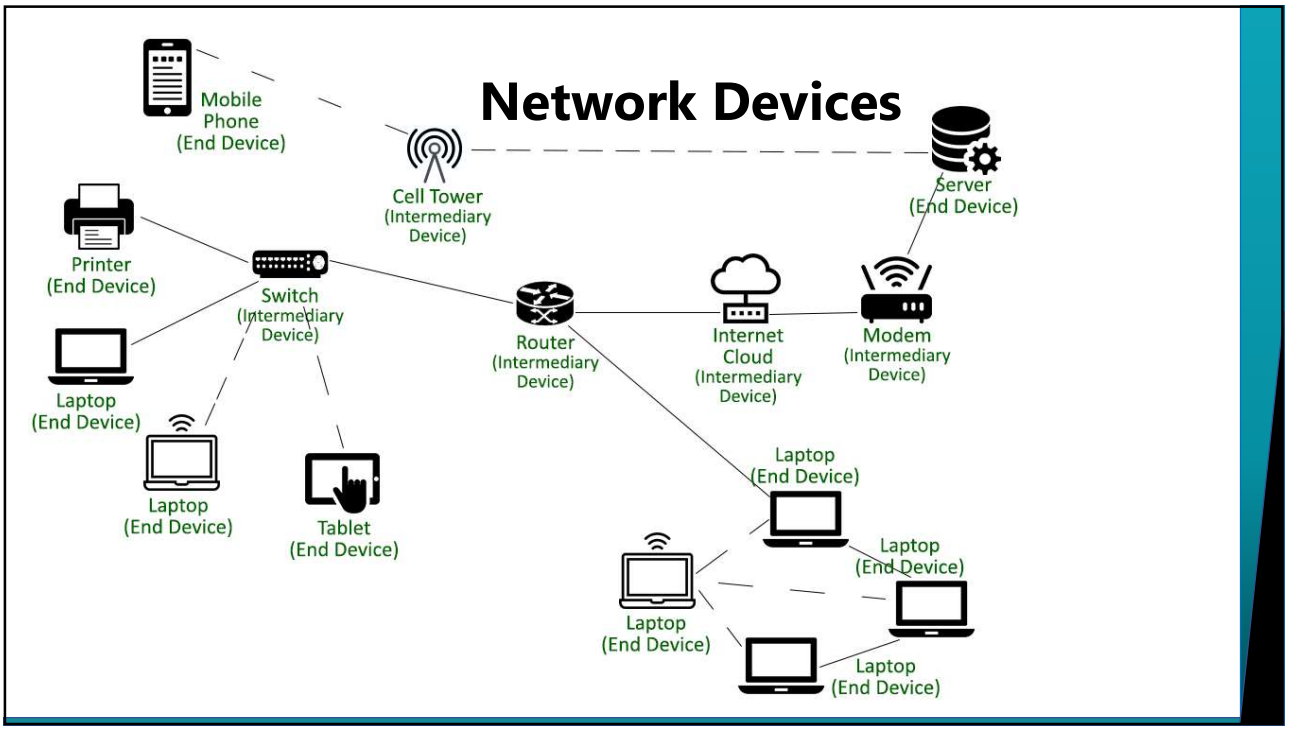


Convergence of functionality: Devices offer a variety of different functionalities

IP Convergence:

Voice and data traffic shares a common network infrastructure

47

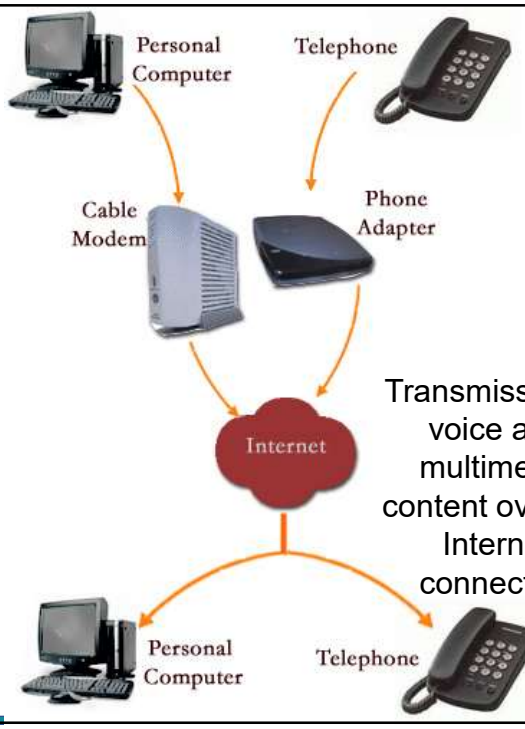


48

Voice over Internet Protocol (VoIP)

Enables,
- Video conferencing
- Business texting
- Call forwarding, …etc

VoIP Software e.g.,
- Dialpad
- CloudTalk
- Skype
- Viber
- Jitsi –free & OpenSource



Transmission of voice and multimedia content over the Internet connection

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Wireless communication

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IS Infrastructure: Data and knowledge

- Infrastructure to support business processes and gather business intelligence
  - Knowledge management systems (KMS)
  - Online transaction processing (OLTP) systems
  - Online analytical processing (OLAP) systems
  - Transactional and analytical processing
    - Operational systems : Interact with customers and run a business in real time
    - Informational systems : support decision making based on stable point-in-time or historical data

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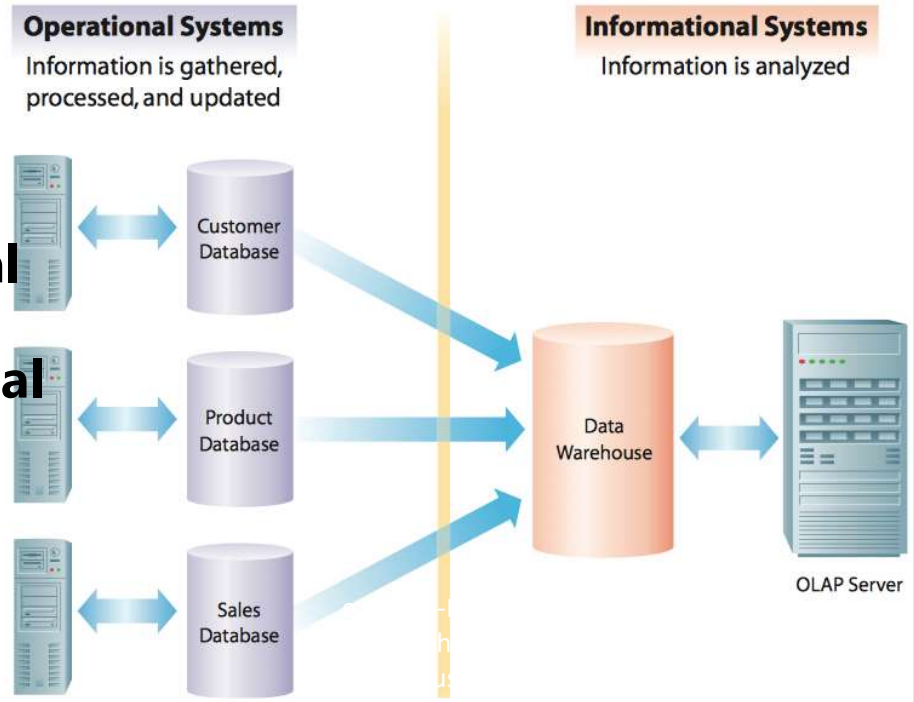
Operational and Informational Systems



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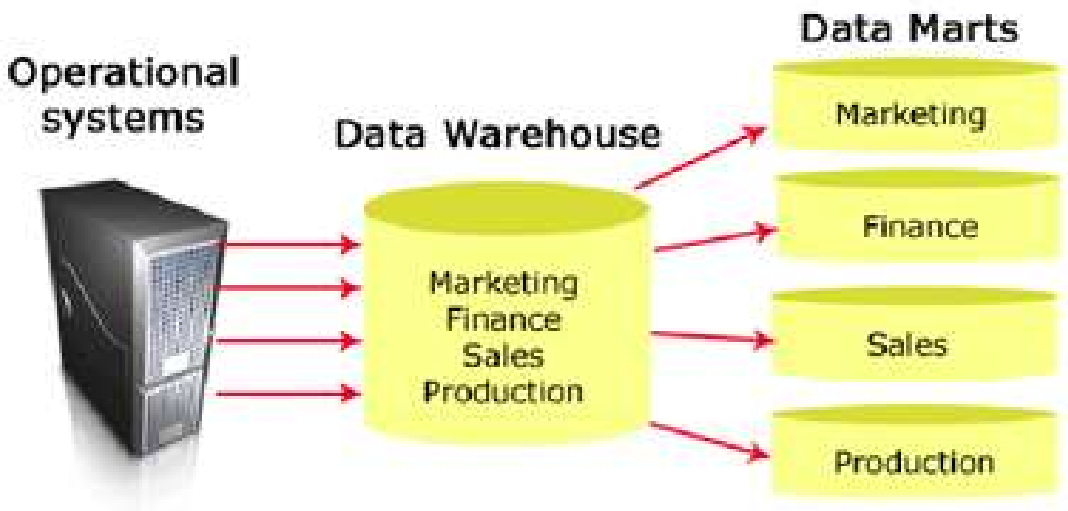
Data Mart



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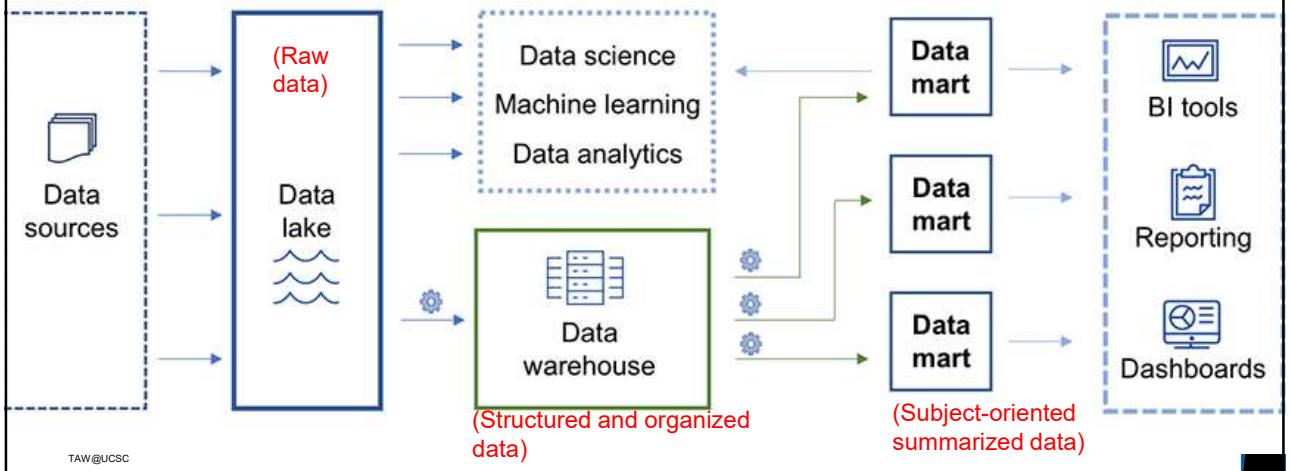
56

Databases



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Activity

- What is the difference between a hub, switch and a router?
- Discuss the difference between a data lake, data mart and a data warehouse.

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IS Infrastructure: Facilities

- Ensure availability –
  - Plan for the worst and
  - Take protective measures
- Securing the facility infrastructure -
  - Use access controls,
  - closed-circuit television (CCTV) monitoring and
  - intrusion detection systems

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IS Infrastructure: Human resource

- Requirement of highly trained workforce
  - Set up shops in areas having skilled workers
  - Provide educational grants
  - Train the staff
  - Implement human resource policies, such as telecommuting, flextime, and creative benefit packages, to attract and retain the best employees

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IS Infrastructure: Services

IS Infrastructure	Service	Example
Hardware	Utility computing	Pay for data processing and storage on an as-needed basis
Software	ASP	Payroll system hosted in an ASP's server
Communication & collaboration	Video conferencing	Use Zoom and pay monthly fee
Data and knowledge	ASP	Data from applications hosted on an ASP's server is stored by the provider
Facilities (e.g. Power, backups, Connectivity, and Security)	Collocation facility	Rent space for their servers in a collocation facility

65

ACTIVITY

1. Discuss what utility computing is.
2. What's the difference between grid computing and cloud computing?
3. What's the difference between data warehouse and data mart?
4. Discuss pros and cons of using services of an ASP.
5. Compare ASP and SaaS.

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66